USE OF ORGANIC MANURE FOR COCONUT

1. Introduction

Continuous organic manure application tends to increase the humus content in the soil, and supply plant nutrients. High humus content improves the water holding capacity, nutrient retention of soils, aeration, structure, microorganism density, microbiological activities. It also maintains the soil pH level at favorable levels to coconut. Therefore organic manures are important sources of plant nutrients, which help in sustaining soil fertility and productivity especially for perennial crops such as coconut.

Application of organic manure would result in an increase of 15-20% in terms of nut yield and 20-25% increase in copra yield per year. If organic manures that are produced in the estate itself such as cow dung and poultry manure are applied, the income will be very much greater as the cost of fertilizer can be reduced.

2. Nutritional value of organic manures

Table 1 and 2 give a list of organic materials and their nutritional values of macronutrients and micro nutrient values of these organic materials.

Table 1: Macronutrient content of locally available organic sources (Dry weight basis).

Organic Source	Nitrogen (N)%	Phosphorus (P)%	Potassium (K)%	Magnesium (Mg)%	Calcium (Ca)%
Goat manure	2.2-3.4	0.3-0.7	1.5-2.5	0.4-0.8	1.5-2.4
Cattle manure	1.2-1.9	0.2-0.5	0.5-1.1	0.5-0.6	1.3-1.8
Poultry manure					
Broiler litter	2.0-2.3	0.6-1.0	1.7-2.0	0.5-0.6	1.0-4.9
Layer litter	1.8-2.4	0.6-1.2	0.6-2.0	0.4-0.7	2.7-5.3
Pig dung	1.0-2.0	0.6-0.9	0.4-0.9	0.4-0.6	1.0-1.5
Farm yard manure	1.2-1.8	0.4-0.6	1.1-1.9	0.5-1.0	0.5-1.1
Compost	1.3-1.7	0.3-0.6	0.4-0.7	0.2-0.5	0.8-2.0
Gliricidia leaves	2.5-3.5	0.1-0.2	1.3-1.7	0.3-0.5	1.0-1.9

Table 2: Micronutrient content of locally available organic sources (Dry weight basis)

Organic Source	Micronutrients ppm (mg/kg)				
	Iron (Fe)	Manganese (Mno)	Cupper (Cu)	Zink (Zn)	Boron (B)
Goat manure	1449-2174	246-505	20-38	112-184	29-66
Cattle manure	690-1518	167-389	24-40	128-183	13-30
Poultry manure					
Broiler litter	723-1565	213-421	27-40	166-271	15-27
Layer litter	1144-2215	287-450	22-38	182-329	12-2
Pig dung	1020-1990	180-207	45-48	186-575	34-13
Farm yard manure	1135-3515	229-66	17-29	83-128	12-27
Compost	2090-4064	8201-50	11-25	75-169	12-20
Gliricidia leaves	212-450	585-156	4-5	57-70	41-69

3. Rate of application of organic manure

Compared to chemical fertilizers, the nutrient content of organic manures are very much low. Therefore, to supply full nutrient requirements of coconut palms, organic manures should be applied in large quantities and some are to be supplemented with chemical fertilizers.The required rates of organic manures with inorganic supplementation for young and adult coconut palms are given in Table 3 and 4 respectively.

Table 3: Rate of application of organic manures and inorganic supplementation for young coconut palms

Source	Time after transplanting				
	6 months	1 years	2 years	3 years	4 years upto bearing (every year)
Cattle manure (moisture 20-30%)	5kg	12kg	16kg	20kg	25kg
+					
Eppawala Rock Phosphate (in wet and intermediate zones)	200g	450g	600g	750g	1000g
or					
Triple Super Phosphate (in dry zone)	85g	200g	270g	340g	420g
+					
Muriate of Potash	30g	200g	250g	325g	400g
+					
Dolomite	250g	250g	250g	250g	250g

Source	Time after transplanting				
	6 months	1 years	2 years	3 years	4 years upto bearing (every year)
Goat manure (moisture 20-30%) +	3kg	7kg	9kg	11kg	13kg
Eppawala Rock Phosphate (in wet and intermediate zones) or	200g	450g	600g	750g	100g
Triple Super Phosphate (in dry zone) +	85g	200g	270g	340g	420g
Muriate of Potash +	50g	120g	150g	190g	225g
Dolomite	250g	250g	250g	250g	250g
Poultry manure (20-30%) +	5kg	12kg	16kg	20kg	25kg
Dolomite	250g	250g	250g	250g	250g
Gliricidia leaves (moisture 20-30%) +	5kg	12kg	16kg	20kg	25kg
Eppawala Rock Phosphate (in wet and intermediate zones) or	275g	650g	825g	1100g	1350g
Triple Super Phosphate (in dry zone) +	120g	285g	370g	470g	580g
Muriate of Potash +	60g	150g	200g	250g	250g
Dolomite	250g	250g	250g	250g	250g

Tale 4 : Amount of application of organic manures with inorganic supplementation for adult coconut palms

Organic manure	Quantity (Kg)	Eppawala Rock Phosphate for wet & intermediate zone/ Triple Supper Phosphate for dry zone (g)	Muriate of Potash (g)	Dolomite (g)
Cattle manure (Moisture 20-30%)	30	-	1250	-
Goat manure (Moisture 20-30%)	25	-	800	-
Poultry manure (Moisture 20-30%)	30	-	750	-
Gliricidia (Moisture 20-30%)	30	525 g (ERP) Wet and intermediate zones or 240 g (TSP) (Dry zone)	1000	500

The poultry manure removed from sheds should be kept outside in heaps for 2-3 months before appling to palms.

4. Method of application

Young palms

In the early stages (up to 1-1 ½ years of age) organic manures should be broadcast up to a distance of 0.3 meters (about 1 feet) from the base of seedling and incorporate with soil. As the palm grows older this area should be gradually extended up to about 1.8 meters (about 6 feet) at the time of flowering.

Adult palms

i. Surface application method is as same as chemical fertilizer application

Organic manure and the inorganic supplements should be broadcast in the entire area of the soil surface around the base of the palm up to a distance of 1.8 meters (about 6 feet). The manure and inorganic supplementation should be incorporated into a depth of 4” – 6” of the soil. Then the area on which manure is applied should be mulched well with weed trash, dried fronds, layer of husks etc.

ii. Organic application in trenches

Cut a circular trench around the palm at 3 feet away from the base of the trunk, 3 feet wide and 0.5 feet deep and apply recommended quantity of organic manure in to the trench, cover it with soil and apply the mulch on it (figure 1) .

figure 1: Trench application of organic fertilizer

5. Time of application

Organic manure should be applied preferably with the onset of rain when the soil is moist. On light sandy soils and on lands, which tend to get water logged, manure and inorganic supplementation should be applied after the heavy rains are over.

6. Frequency of application

Organic manure and inorganic supplements should be applied annually for both young and adult coconut palms. However, alternative application of organic manure is also recommended for estates which are applied with inorganic fertilizer mixtures at least once in 3-4 years in order to improve the quality of the soil.

Soils and Plant Nutrition Division
January 2018